SECTION-I

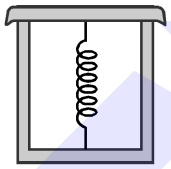
Single Correct Answer Type **7 Q. [3 M (–1)]**

1. At ordinary temperatures, the molecules of a diatomic gas have only translational and rotational kinetic energies. At high temperatures, they may also have vibrational energy. As a result of this compared to lower temperatures, a diatomic gas at higher temperatures will have–

(A) lower molar heat capacity. (B) higher molar heat capacity.

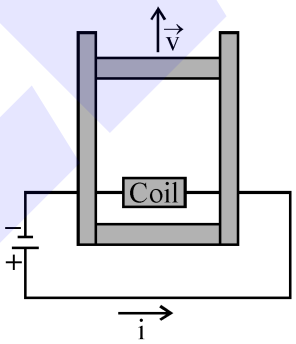
(C) lower isothermal compressibility. (D) higher isothermal compressibility.

2. A container has a lid held in place by a spring as shown in figure. Two moles of ideal mono atomic gas is kept inside the container having cross section area 100 cm^2 at 300 K at atmospheric pressure 10^5 Pa. The spring exerts a force of 1000 N when the lid is in place. The maximum amount of heat that can be given to the gas so that gas does not escape from the container. Assume heat capacity of the spring and container to be negligible. (R = universal gas constant in SI units)



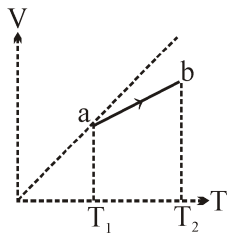
(A) 1200 R (B) 900 R (C) 1500 R (D) 600 R

3. In figure, a resistance coil, wired to an external battery, is placed inside a thermally insulated cylinder fitted with a frictionless piston and containing an ideal diatomic gas. A current i = 200 mA exists in the coil, which has a resistance R = 350 Ω. If the pressure of the gas remains constant, what should be the speed v (in cm/s) of the piston, of mass m = 10 kg ? (Take $p_{atm} = 10^5$Pa, A = 10 cm^2)



(A) 1 (B) 2 (C) 3 (D) 4

4. An ideal gas is heated isobarically from temperature T_1 to T_2 as shown in the V–T diagram. Which of the following statement must be correct?



(A) Heat must be rejected from gas.

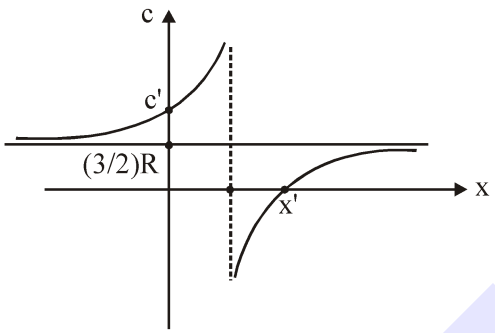
(B) Work must be done on the gas by surrounding

(C) First law of thermodynamics must hold for process a to b

(D) Gas must leak from vessel

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5. One mole of an ideal monatomic gas is taken along the process in which $PV^x = K$. The graph shown represents the variation of molar heat capacity of such a gas with respect to x. The values of c' and x' respectively are given by :-



(A) $\frac{5}{2}R, \frac{5}{2}$ (B) $\frac{5}{2}R, \frac{5}{3}$ (C) $\frac{7}{2}R, \frac{7}{2}$ (D) $\frac{5}{2}R, \frac{7}{5}$

6. The volume thermal expansion coefficient γ for an ideal gas is computed under constant pressure condition. Expressed in terms of its absolute temperature, the correct expression is :-

(A) $\gamma = \frac{1}{2T}$ (B) $\gamma = \frac{1}{\sqrt{2}T}$ (C) $\gamma = \frac{2}{T}$ (D) $\gamma = \frac{1}{T}$

7. What is fraction of molecule below an altitude h in atmosphere? Assume uniform gravitational field, isothermal conditions, mass of a molecule m, Boltzman constant k, temperature T.

(A) $f = e^{(mgh/kT)}$ (B) $f = e^{-(mgh/kT)}$ (C) $f = 1 - e^{-(mgh/kT)}$ (D) $f = 1 - e^{(mgh/kT)}$

Multiple Correct Answer Type **6 Q. [4 M (–1)]**

8. Two moles of $O_2\left(\gamma = \frac{7}{5}\right)$ at temperature T_0 and 3 mole of $CO_2\left(\gamma = \frac{4}{3}\right)$ at temperature $2T_0$ are allowed to mix together in a closed adiabatic vessel. The resulting mixture finally comes in thermal equilibrium. Then

(A) Final temperature of the mixture is $\frac{23T_0}{14}$.

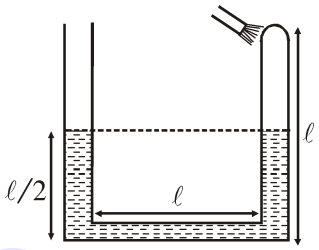
(B) Final temperature of the mixture is $\frac{31T_0}{19}$.

(C) Adiabatic exponent of the mixture formed is $\frac{14}{5}$.

(D) Adiabatic exponent of the mixture formed is $\frac{19}{14}$.

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9. A rectangular narrow U-tube has equal arm lengths and base length, each equal to ℓ. The vertical arms are filled with mercury up to $\ell/2$ and then one end is sealed. By heating the enclosed gas all the mercury is expelled. If atmospheric pressure is P_0, the density of mercury is ρ and cross-sectional area is S, then [Neglect thermal expansion of glass and mercury]



(A) Work done by the gas against the atmospheric pressure is $\frac{5\ell}{2}P_0S$

(B) Work done by the gas against the gravity is $\frac{7}{4}S\rho g\ell^2$

(C) Work done by the gas against the atmospheric pressure is $P_0S\ell$

(D) Word done by the gas against the gravity is $S\rho g\ell^2$

10. A flat plate is moving normal to its plane through a gas under the action of a constant force F. The gas is kept at a very low pressure. The speed of the plate v is much less than the average speed u of the gas molecules. Which of the following options is/are true ? **[JEE-Advance 2017]**

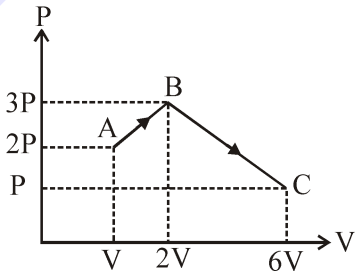
(A) The resistive force experienced by the plate is proportional to v

(B) The pressure difference between the leading and trailing faces of the plate is proportional to uv.

(C) The plate will continue to move with constant non-zero acceleration, at all times

(D) At a later time the external force F balances the resistive force.

11. A monoatomic ideal gas undergoes a process ABC:



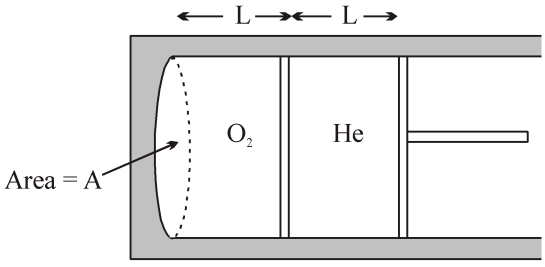
(A) Heat given to the gas is $\frac{33PV}{2}$.

(B) Heat given to the gas is $\frac{35PV}{2}$.

(C) Maximum temperature of the gas during the process is $\frac{8PV}{nR}$.

(D) Maximum temperature of the gas during the process is $\frac{4PV}{nR}$.

12. An insulating cylinder contains equal volumes of He and O_2 separated by a massless freely moving adiabatic piston as shown. The gas is compressed by moving the insulating piston so that volume of He becomes half. Select the correct alternative(s).



(A) Pressure in He chamber will be equal to pressure in O_2 chamber

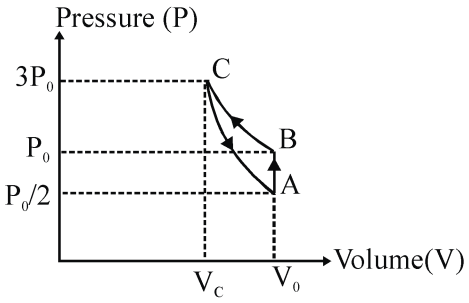
(B) Pressure in He chamber will be less then pressure in O_2 chamber

(C) Volume of He chamber will be equal to volume of O_2 chamber

(D) Volume of O_2 chamber will be $\frac{(LA)}{(2)^{25/21}}$.

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13. One mole of an ideal gas is carried through a thermodynamic cycle as shown in the figure. The cycle consists of an isochoric, an isothermal and an adiabatic processes. The adiabatic exponent of the gas is γ. Choose the correct option(s).



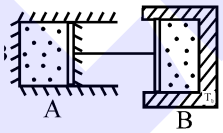
(A) $\gamma = \frac{\ln 6}{\ln 3}$ (B) $\gamma = \frac{\ln 5}{\ln 3}$ (C) BC is adiabatic (D) AC is adiabatic

Linked Comprehension Type **(1 Para × 4Q.) [3 M (-1)]**

(Single Correct Answer Type)

Paragraph for Questions No. 14 to 17 (4 questions)

Two cylinder A and B having piston connected by massless rod (as shown in figure). The cross-sectional area of two cylinders are same & equal to 'S'. The cylinder A contains m gm of an ideal gas at Pressure P & temperature T_0. The cylinder B contain identical gas at same temperature T_0 but has different mass. The piston is held at the state in the position so that volume of gas in cylinder A & cylinder B are same & is equal to V_0.



The walls & piston of cylinder A are thermally insulated, whereas cylinder B is maintained at temperature T_0 reservoir. The whole system is in vacuum. Now the piston is slowly released and it moves towards left & mechanical equilibrium is reached at the state when the volume of gas in cylinder A becomes $\frac{V_0}{2}$.

Then (here γ for gas = 1.5)

14. The mass of gas in cylinder B

(A) $2\sqrt{2}$ m (B) $3\sqrt{2}$ m (C) $\sqrt{2}$ m (D) none

15. The change in internal energy of gas in cylinder A

(A) $(\sqrt{2}-1)PV_0$ (B) $2(\sqrt{2}-1)PV_0$ (C) $\frac{PV_0}{(\sqrt{2}-1)}$ (D) none

16. If work done by gas in cylinder B is W_B & work done by gas in cylinder A is W_A then

(A) $W_A = -W_B$ (B) $|W_A| > |W_B|$ (C) $|W_A| < |W_B|$ (D) we can't say anything

17. What will be the compressive force in connecting rod at equilibrium

(A) PS (B) $\sqrt{2}$ PS (C) $2^{3/2}$ PS (D) none

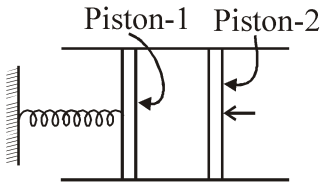
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SECTION-II

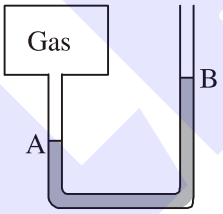
Numerical Answer Type Question **3 Q. [3(0)]**

(upto second decimal place)

18. A long container has air enclosed inside at room temperature and atmospheric pressure (10^5 pa). It has a volume of 20,000 cc. The area of cross section is 100 cm^2 and force constant of spring is k_{spring} = 1000N/m. We push the right piston isothermally and slowly till it reaches the original position of the left piston which is movable. What is the final length of air column in cm. Assume that spring is initially relaxed.



19. Figure shows an ideal gas. Its pressure, volume & temperature are P_0, V_0 & T_0 respectively. Thin U-tube contains mercury. It is seen that when 20 cm of extra mercury is added in limb B of U-tube and temperature of gas is doubled, level of A is maintained at its position. Consider surrounding to be vacuum. Find the initial pressure of gas (in cm of Hg.)



20. A balloon is filled with helium at the atmospheric pressure P_0. The volume of the balloon is V_0. The balloon is made of the material of mass m and density ρ. After being released, the balloon bursts at an altitude where the atmospheric pressure is (P_0/2). Immediately before bursting, the balloon has a volume of 1.25V_0. The maximum stress that the balloon material can withstand is given by αP_0, find the value of α. Assume that the temperature of helium remains constant, the balloon remains spherical, and the density of the material remains virtually constant.[Take : V_0 = 8 m^3, ρ = 1500 kg/m^3, m = 18 kg]

SECTION-III

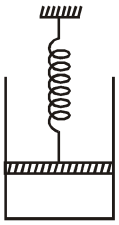
Numerical Grid Type (Ranging from 0 to 9) **3 Q. [4 M (0)]**

21. One mole of an ideal monoatomic gas is taken from state A to state B through the process $P = \frac{3}{2}T^{1/2}$. It is found that its temperature increases by 115.2 K in this process. Now it is taken from state B to C through a process for which internal energy is related to volume as $U = \frac{1}{2}V^{1/2}$. The volume at B is 100 m^3 and at C it is 1600 m^3, then the total work performed by the gas is $\alpha \times 10^2$ J. Fill α in OMR sheet.

(Use R = $\frac{25}{3}$ J/mol-K)

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22. An ideal monoatomic gas is enclosed in a vertical cylinder of cross section area A and length ℓ, under frictionless piston connected to spring of spring constant K. Atmosphere pressure is such that in absence of gas under the piston, equilibrium of piston is achieved when piston just touches the bottom of the cylinder and spring is in stretched configuation. Q amount of heat is supplied to gas slowly to move piston to the upper edge. Initially volume occupied by gas is $\frac{1}{3}$rd of the volume of cylinder. If $Q = 2\lambda\frac{K\ell^2}{9}$, find λ, (through the motion of the piston spring is in stretched configuration)

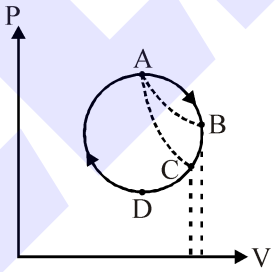


23. A solid object has density ρ, mass M, and coefficient of linear expansion α. At pressure P the heat capacities C_p (at constant pressure) and C_v (at constant volume) are related by $C_p - C_v = \eta(\alpha MP/\rho)$. Fill the value of η in your OMR sheet.

SECTION-IV

Matrix Match Type (4 × 5) **2 Q. [8 M (for each entry +2(0)]**

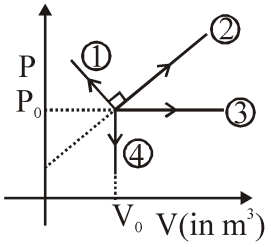
24. For an ideal gas a process PV diagram is a circle. An adiabatic from A passes through C. An isotherm from A passes through B. We take a part of the circular cyclic process. Comment on the sign of the quantity of column-I.



	Column-I		**Column-II**
(A)	Heat given to the gas in going from A to C along circle	(P)	Positive
(B)	Heat given to the gas in going from B to C along circle	(Q)	Negative
(C)	Heat given to the gas in going from C to D along circle	(R)	Zero
		(S)	can't be said

25. A monatomic ideal gas at pressure P_0 & volume V_0 can be made to undergo 4 different process on PV diagram as shown. For each of the processes, match with entries in column-II. (Symbols have their usual meanings).

	Column-I		**Column-II**
(A)	Process–1	(P)	ΔQ + ve
(B)	Process-2	(Q)	ΔU + ve
(C)	Process-3	(R)	W –ve
(D)	Process-4	(S)	Temperature decreases
		(T)	ΔQ – ve



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Subjective Type **5 Q. [4 M (0)]**

26. A movable heavy piston is supported by a spring inside a vertical cylindrical container as shown. When all air is pumped out of the container, the piston is in equilibrium as shown in Diagram 1, with only a tiny gap between the piston and the bottom of the container. When a portion of gas at temperature T is introduced under the piston, the later rises to a height h as shown in Diagram 2. What would be the height of the piston above the bottom of the container if the gas is then heated to a temperature 2T ? Assume that the piston moves without friction and that the spring obeys Hooke's law.

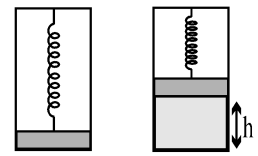
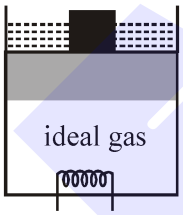


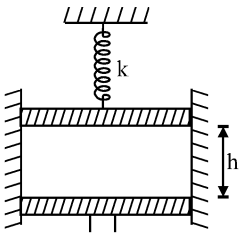
Diagram 1 Diagram 2

27. A block of ice of mass 10 gm is in thermal equilibrium in 10 gm of water both are in contact of a conducting movable massless piston on a cylinder containing 3 moles of an ideal diatomic gas. In thermal equilibrium with water, the walls of cylinder are adiabatic and heat lost to surrounding is negligible. The gas is heated slowly by a heater. Find,

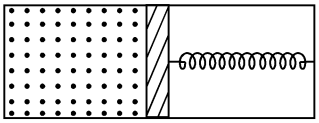


(a) work done by the gas by the time ice melts completely,
(b) net heat absorbed by the gas by the time ice melts.
(c) net heat supplied by the heater by the time water starts boiling.
[Given $L_f = 80$ cal/gm, $R = 25/3$ J/mole – k, $S_W = 1$ cal/gm, mechanical equivalent of heat J = 4.2 Joules/cal]

28. An ideal gas at NTP is enclosed in a adiabatic vertical cylinder having area of cross section A = 27cm^2, between two light movable pistons as shown in the figure. Spring with force constant k = 3700 N/m is in a relaxed state initially. Now the lower piston is moved upwards a height h/2, h being the initial length of gas column. It is observed that the upper piston moves up by a distance h/16. Find h taking γ for the gas to be 1.5. Also find the final temperature of the gas.

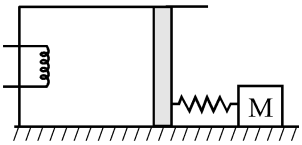


29. A thermally insulated vessel is divided into two equal parts by a heat-insulating piston which can move in the vessel without the friction. The left part of the vessel contains one mole of an ideal monatomic gas, & the right part is empty. The piston is connected to the right wall of the vessel through a spring whose length in free state is equal to the length of the vessel as shown in the figure. Determine the heat capacity C of the system, neglecting the heat capacities of the vessel, piston and spring.



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30. An adiabatic cylinder has 8 gram of helium. A light smooth adiabatic piston is connected to a light spring of force constant 300 N/m. The other end of the spring is connected with a block of mass 1 kg kept on a rough horizontal surface of coefficient of friction $\mu = 0.3$. Area of cross-section of cylinder is $A = 25\ cm^2$. Initially the spring is in a relaxed position and the temperature of the gas is 400 K. The gas is heated slowly for some time by means of an electric heater so as to bring the block M on the verge of motion. Take $P_{atm} = 10^5\ N/m^2$. Find



(a) the work done by the gas
(b) the final temperature
(c) heat supplied by the heater

KTG & THERMODYNAIMCS FIRST LAW — ANSWER KEY

SECTION-I

Single Correct Answer Type — **7 Q. [3 M (–1)]**

1. Ans. (B) **2. Ans. (B)** **3. Ans. (B)** **4. Ans. (D)** **5. Ans. (B)**
6. Ans. (D) **7. Ans. (C)**

Multiple Correct Answer Type — **6 Q. [4 M (–1)]**

8. Ans. (A, D) **9. Ans. (A,B)** **10. Ans. (A,B,D)** **11. Ans. (A,C)** **12. Ans. (A, D)**
13. Ans. (AD)

Linked Comprehension Type — **(1 Para × 4Q.) [3 M (-1)]**

(Single Correct Answer Type)

14. Ans. (B) **15. Ans. (B)** **16. Ans. (C)** **17. Ans. (C)**

SECTION-II

Numerical Answer Type Question — **3 Q. [3(0)]**

(upto second decimal place)

18. Ans. 100 **19. Ans. 20** **20. Ans. 375**

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) — **3 Q. [4 M (0)]**

21. Ans. 5 **22. Ans. 8** **23. Ans. 3**

SECTION-IV

Matrix Match Type (4 × 5) — **2 Q. [8 M (for each entry +2(0)]**

24. Ans. (A) P; (B) Q; (C) Q **25. Ans. (A) R; (B) P, Q; (C) P,Q; (D) S,T**

Subjective Type — **5 Q. [4 M (0)]**

26. Ans. $h' = \sqrt{2}\,h$ **27. Ans. 0, 0, 14650/3 cal** **28. Ans.** 1.6 m, 364 K
29. Ans. C = 2R **30. Ans.** (a) 2.515, (b) 404.8 K, (c) 122.179 J

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